H3 – Daniyal Shaven team 10

Problem: Minimize $6y_1 + 7y_2 + 5y_3$

Subject to: (T_1) $y_1 + y_2 \geq 1$

(T_2) $y_2 + y_3 \geq 1$

(T_3) $y_1 + y_3 \geq 1$

$0 \leq y_i \leq 1$ $\quad b = (1, 1, 1)$

reduced cost formula:

$$r_j = c_j - \sum_i \pi_i A_{ij}$$

- y_1 appears in rows T1 and T3
- y_2 appears in rows T1 and T2
- y_3 appears in rows T2 and T3

(a) Given $x = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, $\pi = (4, 3, 2)$

Objective function:

$$c^T x = \frac{6}{2} + \frac{7}{2} + \frac{5}{2} = 9$$

Reduced costs:

- $r_1 = 6 - (4 + 2) = 0$

$r_2 = 7 - (4 + 3) = 0$

$r_3 = 5 - (3 + 2) = 0$

a) Primal feasibility: Each constraint has a left-hand side equal to 1, so all constraints are satisfied. Also, $0 \le y_j \le 1$. Violation = 0

b) Sign conditions: Since the constraints are of type ≥, we need $\pi_i \ge 0$. Since $4, 3, 2 \ge 0$, violation = 0

c) Complementary slackness: The slack of every constraint is 0, so $\pi_i \cdot 0 = 0$. Violation = 0

d) Reduced costs: All variables are strictly between 0 and 1, so we require $r_j = 0$. Since $r = (0, 0, 0)$, violation = 0

e) Strong duality:

$$b^T \pi = 4 + 3 + 2 = 9$$

$$\sum_j \min(0, r_j) = 0$$

$$c^T x - b^T \pi = 9 - 9 = 0 \quad \text{Violation} = 0$$

Conclusion: All five conditions are satisfied.

Therefore, x is optimal and $\underline{Z_{LP} = 9}$

(b) Given $\pi^1 = (5, 2, 1)$

reduced costs: $r_1 = 6 - (5 + 1) = 0$

$r_2 = 7 - (5 + 2) = 0$

$r_3 = 5 - (2 + 1) = 2$

a) Primal feasibility: Violation = 0

b) Sign condition: $5, 2, 1 \geq 0$ Violation = 0

c) Complementary slackness: All constraint slacks are 0. Violation = 0

d) Reduced costs: since $y_3 = \frac{1}{2}$ is strictly between 0 and 1, we require $r_3 = 0$. However, $r_3 = 2$. Violation = 2. Condition not satisfied.

e) strong duality:

$$b^T \pi' = 5 + 2 + 1 = 8$$

$$\sum \min(0, r_j) = 0$$

$$c^T x - b^T \pi' = 9 - 8 = 1$$

The duality gap is 1. Condition not satisfied.

(c) Why does π' still provide a lower bound?

Since $\pi' \geq 0$ and $r = (0, 0, 2) \geq 0$, π' is dual-feasible.

For any primal-feasible y,

$$c^T y = \sum_j r_j y_j + \pi'^T A y$$

Because $r_j \geq 0$, $y_j \geq 0$, $Ay \geq b$, and $\pi' \geq 0$, we obtain

$$c^T y \geq \pi'^T b = 8$$ Therefore, $\underline{z_{LP} \geq 8}$

The best lower bound obtained from $\pi = (4, 3, 2)$ is 8.

The gap between the optimal value is $\boxed{9 - 8 = 1}$

done 22.10.20